EXP NO: 13

Setting up a local Area Network

Aim:

To design, set up and configure a local area network connecting 4 computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command and demonstrate data sharing across the network

IP configuration table:

Device	IP Address	Subnet Mask	connected to
PC0	125.1.1.0	255.0.0.0	Switch Fa 0/1
PC1	125.1.1.1	255.0.0.0	Switch Fa0/2
PC2	125.1.1.2	255.0.0.0	Switch Fa0/3
PC3	125.1.1.3	255.0.0.0	Switch Fa 0/4
Switch	125.1.1.4	255.0.0.0	VLAN 1 Interface

Implementation:

Step 1: open cisco packet tracer and place switch, 4 PCs and one server in the workspace

Step 2: connect all PC's and the server to the switch using copper - straight through cables.

Step 3: configure all IP addresses of all PCs and the server so that they belong to the same network

Step 4: verify that all cable connections are active

Step 5: Test Network connectivity by opening

the command prompt on a pc and using the ping command to communicate with another PC or the server.

Step6: Enable the HTTP service on the Server by selecting

server -> Services -> HTTP -> ON

Step 7: Open the web browser on a Pc and enter the servers IP address to access the server's default web page

Step8: Switch from realtime mode to Simulation mode in packet tracer

Step 9: Send a ping or web request from a PC to the server and use the Captured Forward or Auto capture/play option to observe the packet traveling through the switch

Step 10: Verify that Successful Communication takes place b/w the PC's and the servers, demonstrating proper LAN operation and data flowing

Result:

A Local Area Network (LAN) was successfully established using PCs, Switch, and a server, Communication b/w devices verified through ping and web access, and packet flow was observed Successfully in simulation mode